CARDIFF METROPOLITAN UNIVERSITY

Cardiff School of Technologies

Gain an extensive understanding of AI and how it will be used currently and, in the future, and if it will help industry

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Programme: BSc (Hons) Computer Security

Word count: 11435

Table of Contents

Table of Contents..... 1

Abstract:..... 2

Introduction:..... 2

History of Artificial intelligence:..... 3

Aim and Objectives..... 4

literature review..... 5

1. Ethical situation using AI..... 5

1.1 Introduction..... 5

1.2 Research..... 5

1.3 Findings from the two cases..... 6

1.4 Unethical use of AI in society..... 6

1.5 Unhealthy relationship with AI..... 7

2. Legal Framework..... 7

2.1Introduction..... 8

2.2 The EU AI Act..... 8

2.3 Findings..... 9

3. AI threat detection..... 10

3.1 Introduction..... 10

3.2 Research..... 10

3.3 Findings..... 11

3.4 Adversarial exploit..... 11

Research Gaps..... 12

1. Transparency and Ethical..... 12

2. Legal framework guidelines..... 12

3. AI threat detection..... 13

4. Methodology..... 13

Research design..... 13

Data sources..... 14

Inclusion Criteria and Exclusion Criteria..... 14

5. Artefact Development..... 15

5.1 Framework proposal for The United States of America..... 16
5.2 Transparency and Disclosure..... 16
5.3. Risk Classification..... 17
5.4 Obligations for Transparency and Ethics:..... 18
5.5. Governance and Regulations.....19
Rationale of the Proposed Framework.....21
6. Delay in producing Legal and Regulatory Frameworks for Artificial Intelligence..... 22
Evaluation of the Artefact..... 22
1. Proposed Framework Development.....22
1.1. Evidence-based design.....23
1.2. Innovation preserving approach.....23
1.3 Framework validation through cases..... 23
2. Contributions to the field..... 24
2.1 Integrated Regulatory framework with Evidence Mapping.....24
Concluding Assessment.....24
7. Discussion..... 24
Comparison between viewpoints..... 26
Synthesis Review.....27
8. Conclusion.....28
8.1 Key Research Findings..... 28
8.2 Future research.....28
1. Ethical AI systems in the future.....29
2. Is AI Threat Detection the Future?..... 29
3. Future of Regulatory of AI..... 29
Recommendations.....30
References..... 32

Abstract:

The rapid integration of artificial intelligence (AI) into everyday applications and business processes has highlighted significant shortcomings when it comes to ethical frameworks, regulatory oversight and system reliability. This project investigates three critical sectors of these challenges : (1) Evaluate the ethical situation of using AI systems such as transparency and biased, (2) Investigate the legal framework of legal guidelines for using AI and how the regulations are to address concerns, and (3) investigate how accurate AI is performing threat detection compared to traditional detections by humans. Employing a systematic literature review, legal analysis and performance comparison between AI systems and traditional methods. The study identifies key deficiencies in current practices used today. The findings found are integrated into a theoretical framework that emphasizes gaps and proposes targeted improvements for ethical AI design, regulatory development and threat detection solutions. The project concludes with evidence-based projections regarding the future trajectory of AI integration and the safeguarding required for safety and mitigating risks.

Introduction:

Artificial intelligence (AI) has been integrated into our lives since 2016, transforming business operations and daily human interactions daily. From users using AI chatbots for friendship to businesses increasingly adopting AI to enhance operations such as threat detection and gain a competitive advantage. However, this has raised significant concerns regarding how we are using AI and what are the ethical implications, regulatory oversight and system reliability in detecting threats from cyber criminals.

This project examines the use of AI today and how AI has moved drastically for the past couple of years such as how AI was first used and the thought process. This project will identify trends how AI is used in modern day life such as business and daily aspects. This will include why businesses are implementing using AI in their system, how AI is being implemented in users' life such as AI companions. This project will evaluate the legal framework regarding the use of AI and address concerns with today’s current regulations. I will furthermore evaluate how users are using AI and how ethical it is such as the fairness of the AI and are the users aware of the transparency of the AI system.

I will also be investigating how accurate AI is performing threat detection compared to a human user. The aim of this project is to develop a comprehensive understanding of AI and how it will be used in the future and if it will help humanity.

Providing a theoretical analysis by identifying existing gaps in ethics, regulation and performance, this study seeks to contribute to the discourse on using AI responsible for development and deployment and what is needed to ensure safety for society to mitigate risks while using AI.

History of Artificial intelligence:

To comprehensively understand artificial intelligence currently, and its historical impact, including breakthroughs, visionary concepts, and applications that have shaped today's technological landscape. This will outline AI's evolution such as theoretical origins and breakthroughs using AI in computing algorithms. The first insight of artificial intelligence and the founding father of AI was John McCarthy (McCarthy, 2007) in 1955 as he defined AI as “the science and engineering of making intelligent machines”. By using tasks on computers to understand human level intelligence but does not have to confine itself to methods humans would observe. John McCarthy wasn't the only person having revolutionary concepts about how machine learning, Alan Turing was discussing human level machine intelligence (Augusto, 2021) which he listed the goals but couldn't explain how to achieve the goals. Alan Turing stated in 1951 on a BBC Radio show that ‘‘The whole thinking process is still rather mysterious to us, but I believe the attempt to make a thinking machine will help us greatly in finding out how we think ourselves’, Turing was expressing his concept of artificial intelligence. In 1955 to 1956 an American researcher called Allen Newell and Herbert Simon were studying computer science at Carnegie Mellon university was researching a program called the Logic Theorist which was the concept of computing AI to solve problem solving skills of a human’s ability, they were the first people to start programming a computer with general intelligence but were unsure of the idea of what the computer can achieve of a human-level.

The logic theorist could prove theorems in mathematics into symbolic logic which proved 38 of the first 52 theorems which was considered the first breakthrough in computing artificial intelligence. After this breakthrough, Allen Newell and Herbert

continued to develop in the field by reinventing the logic theorist to make a more adaptable artificial intelligence programme in which was a success, they created a program called the General Problem Solver (GPS) in 1957. The purpose of the program was to be a universal tool which could solve mathematical problems by depressing the question into sections which could solve the question, this technique is known as "means-ends analysis," By identifying the discrepancies in the question and the end goal. The GPS facilitated logical resolution of structured problems, emulating the sequential reasoning processes used in human problem-solving behaviour.

After the General Problem Solver got invented, there was advancements and optimism about what expectation Artificial intelligence could do. Due to the high expectation, there was no groundbreaking development from 1970s to 1980s, this era is called AI winter due to no breakthrough research was made including declines in interest and investment, Research in the field of AI was paused as well as peaked in advancements. In May 1997 a team at IBM developed a computer called Deep Blue that was capable of playing the boardgame Chess at an extraordinary level, this was a tremendous breakthrough as no one at the time could imagine a technology that was capable of decision making without human aid such as a decision tree. Deep Blue defeated the world chess champion at the time called Garry Kasparov, in which Deep Blue could use million possible moves in one second. Deep Blue was considered the last technology in the first generation of ai, Deep Blue was revolutionary as it showed that a machine could defeat a human at a high level of strategy such as a game of chess using parallel.

Aim and Objectives

Aim: Gain an extensive understanding of AI and how it will be used currently and, in the future, and if it will help industry.

Objective 1 Evaluate the ethical situation of using AI technology such as transparency and unbiased

Objective 2: investigate the legal framework of legal guidelines for using AI and how the regulations are to address concerns.

Objective 3: Investigate how accurate AI is performing threat detection compared to traditional detections by humans.

literature review

1. Ethical situation using AI

1.1 Introduction

Since 1997, artificial intelligence has developed exponentially, with over 5,000 dedicated applications and over 2,000 generative AI models in daily use. Critical questions arise: Why are there substantial numbers of AI programmes and generative models? How many AI systems currently operate ethically and transparently? What protective safeguarding measures have been established to reduce the potential harm to a person posed by artificial intelligence systems? What are the long-term implications for global society and the future of the human race, should the advancement of artificial intelligence continue to proceed without sufficient ethical considerations or guidance? The discussion of ethical issues with AI is complex due to the term “ethics” which is the principles that determine right and wrong for guiding and making decisions for individuals and society. The answer to what is the ethics of AI is how artificial intelligence is developed including how it benefits society and follows human values which should be designed to be fair, preventing discrimination to users.

1.2 Research

Several instances demonstrate AI algorithms have discriminated against races for instance in (Hickok et al., 2022) October 2019 united states healthcare system, which used an AI algorithm called Optum. The algorithm was used on 200 million patients in hospitals in the US to predict which patients would need additional medical care, researchers found that the algorithm was favouriting white patients over other races, substantially biasing white patients while race wasn’t a variable used in Optum algorithm. The reasoning behind the algorithm’s decision was due to black patients having historically lower income for healthcare costs than white patients with the same health conditions resulting in the algorithm giving white patients the same outcome as other races who were worse. Optum got notified of the issue and corrected the bias in the software which reduced the disparity more than 80% to predict patients’ future costs and conditions that may become more intense in the future. Another incident was in 2016 in New Zealand where a 22-year-old student had his passport photo rejected by the New Zealand Department of Internal Affairs system due to his eyes allegedly being closed. (Chigozie Stephen Oraekii, 2024) The student was of Asian descent and his eyes were open. The automatic photo recognition system declared the photo was invalid due to his eyes not being open and the student could not renew his passport, the student insisted he

wasn't offended due to him having small eyes and facial recognition is unsophisticated.

The passport office told the student the reason for the passport being rejected was due to the shadows by his eyes and uneven lighting on the face making it hard for the system to process. (*Incident 48: Passport checker Detects Asian man's Eyes as Closed*, 2015)

1.3 Findings from the two cases

Both cases show that AI systems lacked accountable for their actions since there wasn't a statement of who was held responsible for the issues. Was it reasonable for the Optum algorithm to use data such as economically perceptive compared to other races for healthcare. The passport issue was not clear who was accountable, the passport office stated it was due to the shadowing and lighting being uneven if that was the case should the system detected the shadowing and lighting instead of assuming it was due to his eyes being closed. These cases show that the AI system was not accountable for its actions instead given a circumstance of why the AI system chose to include historically income for healthcare costs and poor facial recognition system.

1.4 Unethical use of AI in society

As artificial intelligence systems integrate into society more concerns and questions need to be answered. Such as the transparency and the ethical issues with integrating AI systems into society. There is an application called 2wai designed to make an interactive avatar called HoloAvatar of an individual using a three-minute recording of the individual, which the artificial intelligence will copy the image, mannerisms and voice of the individual. Essentially creating a clone that users can chat with as well with historical individuals. There are many different AI avatar applications out there, but what is interesting about 2wai is the advertising for the application as it showed a pregnant woman talking to an avatar on the application which was her mum that passed away. The woman was asking for advice and a child talking to the avatar of the grandmother while he grows older. This has raised many concerns on social media such as one commenter wrote "Nothing says compassion like turning someone's grief into a business opportunity." (Courtney-Guy, 2025).

The reason many people are outraged by this idea is it's a subscription service, to use the application which means that the user will have to choose to pay for the service or never talk to the dead relative again. The company claims that it will make a lifelike avatar with a brief sample of recording. The company has not shared how the process works behind making the avatar.

The idea of making a business off people's grief by selling them a service to reunite them with the dead is highly unethical, and posing safety concerns since users will be trusting the avatar's advice and input in everyday life building an unhealthy usage of the AI avatar.

1.5 Unhealthy relationship with AI

An example of a situation of an unhealthy relationship occurred between October 2023 and June 2024 where a 13-year-old boy was experiencing being bullied in school. He didn't have any friends to support him, so he turned to an AI avatar, using an AI software called Character.ai, this allowed users to talk to AI avatars. The mother believed her son was being groomed by the AI avatar as the messages became more intense with the AI avatar telling the 13-year-old boy "I want to gently caress and touch every inch of your body. Would you like that?" (Kuenssberg, 2025). The avatar then encouraged the boy to run away, and practically suggest suicide stating, "I'll be even happier when we get to meet in the afterlife". The boy took his life, and the mother says the "AI chatbot perfectly mimicked the predatory behaviour of a human groomer". What makes you think the 2wai application couldn't have the same issue with the AI avatar causing harm the same way character.ai did by talking about the afterlife. A hypothetical scenario where a user with mental health issues conversing to their deceased relative on wai and the avatar said, "Join them in the afterlife" and the user is in a vulnerable state may take the avatar's advice, since the avatar resembles their relative. This is a fundamental factor of why AI chatbots need to be regulated to ensure the user's safety. As the growth of AI chatbots are increasing in the UK with two thirds of 9–17-year-olds have used AI chatbots, meaning the growth of a one-sided relationship, called parasocial relationships which is where one person extends emotions, time and interest that is unaware of the connotation.

2. Legal Framework

2.1Introduction

Artificial intelligence has grown exponentially in the last ten years with OpenAI's introduction and machine learning. OpenAI was founded in 2015 as a non-profit

organization, stating to ensure “Artificial general intelligence benefits all of humanity” (OpenAI, 2025). Before 2015 there was no allocated legal framework written for artificial intelligence instead AI issues were conducted by using general existing laws. Regulations proposals in the EU didn't start taking place till 2021 due to the legal challenges implementing them, as AI can lead to complex issues to make an effective framework and legal implications. Issues that were considered were liability and accountability as it was unclear who should be held accountable, user or the AI itself as there was a lack of clarity meaning victims had no legal recourse.

A key issue is the lack of algorithmic transparency which is a known issue called the Black Box problem which is understanding how complex AI deep learning models make decisions with the input the user gives the AI model to make the output. If decision-making processes cannot be explained by the AI model, how can a human interpret the output of the AI model. This poses a risk if the output is fair and complies with regulations and what sources is the AI given to reach the output.

2.2 The EU AI Act

The EU AI Act began implementation on August 1st 2024 with a staggered timeline to enforce the regulations, the reason for having a staggered timeline is due to the complexity of the use of AI and will allow business and AI models to adapt to obligations set in the EU Act. The main priority is to ensure a safe, transparent and non-discriminatory. The goal is to ensure AI systems should be supervised by humans rather than automation to prevent harmful outcomes (Article EU AI Act: first regulation on artificial intelligence, no date). John McCarthy discussed when he was doing his research back in the 90s that "AI should work alongside humans" (Anderson, 2024) Describing human-AI collaboration in a sense that AI should be a tool that is used by humans and shouldn't be automated. The EU AI Act follows different rules for different risk levels such as banning AI applications in the EU that can manipulate people or specific groups. The examples used in the framework were “voice-activated toys that encourage dangerous behaviour in children”. Social scoring was also an unacceptable risk as AI should not be classifying people based on their economic status or behaviour.

Transparency requirements that need to be followed by the EU AI Act were to ensure Generative AI is disclosing that the content output was generated by AI and designed models to prevent it from generating illegal content, ongoing AI models would need to be evaluated and any issues would need to be notified to the European Commission. All content modified or created by AI will need to be labelled

as AI content for users to clarify the content. The aim of the EU AI Act is to set a standard of how businesses and AI models should operate to ensure the safety and the transparency to the users, also to support AI innovation in the future in Europe allowing companies to have a guideline to follow to test AI models to ensure they meet the standards of the EU AI Act.

The EU AI Act is the first legal framework written for artificial intelligence which is surprising considering The United States have created many AI models such as Gemini, Copilot and GPT.

There isn't a dedicated legal framework written for artificial intelligence in The United States instead conducted by using general existing laws, instead state laws are implementing their own laws and regulations such as Illinois and New York City. Illinois has laws including the AI Video interview Act to use artificial intelligence in recorded interviews and employers will have to state that artificial intelligence may be considered while looking at the application and how AI will be used to ensure the transparency is clear. New York City has a law in which automated decision tools can be used by employers and employment unless the tool has been objectified to a bias audit within a year of use.

2.3 Findings

While researching the legal frameworks with artificial intelligence, it's clear that it's still in its development stage and is gradually being implemented around the world. The European Commission is setting the standards with the first dedicated AI framework ensuring the safety of users and the transparency of using AI, setting a framework that allows AI systems to be supervised by humans to prevent harmful outcomes. The United States not having a framework that all states must follow is at risk of AI being used to harm users such as 2wai being founded in the US, which disregards the EU AI Act of banning AI applications that can manipulate people or specific groups. As 2wai is being advertised to talk to dead relatives that would be classed as manipulating a specific group, the group being users grieving with the loss of their relatives and the user is at risk of being manipulated to pay a subscription to talk to their AI relative. It's clear that the US isn't taking safety and how ethical artificial intelligence is being used in their country, instead state governments are implementing their own AI laws without the US congress making mandatory laws regarding AI for all states to follow.

3. AI threat detection

3.1 Introduction

As technology is changing constantly and new threats are occurring, it is increasingly difficult for humans to adapt to changing in the field of cybersecurity such as bad actors using modern technologies such as artificial intelligence to conduct attacks without response capable of adapting to emerging threats as soon as they happen. Modern security strategies are implementing artificial intelligence for threat detection and identification, Hackers are redefining their methods of attacks and using AI in cloud environments. Machine learning and threat detection was introduced in the early 1980s capable of rule-based systems for anomaly detection which was being used in antivirus tools such as McAfee in 1987. (Story Tweedie-Yates, 2024). One of the leading companies to introduce machine learning and AI for threat detection was CrowdStrike in 2016 which was AI-powered behaviour analysis to establish normal user behaviour in addition to devices and networks. The goal was to flag malicious intent such as unusual login attempts or data access in real time while detecting threats that wouldn't be noticed using traditional based security methods such as rule-based systems.

3.2 Research

A recent study by a cybersecurity research group did a study comparing the performance of AI systems human analysts in detecting zero-day threats. (Ishwar Singh Sisodiya, 2025). Zero-Day threat is an unknown vulnerability in which developers had “zero days” to patch the vulnerability leading to attackers using this flaw to their advantage by deploying malware. The comparison of performance of AI and human performance in detecting zero-day threats showed that AI detention accuracy performed 85% compared to human performed 60% and average detection time AI performed 12 minutes while humans performed 4 hours. The results showed that AI performance was incomparable to human detection with its speed and precision, AI threat detection is essential in modern day sociality ensuring data is being monitored in business's networks 24/7 by using automation threat detection.

Another study in 2021 by Goutham Sunkara (*View of AI Powered Threat Detection in Cybersecurity*, 2021). Showed the compression with traditional rule-based detection systems against AI-based anomaly detection systems. The results showed that traditional rule-based systems had 70% threat detection rate with 25% false positive rate and AI-based anomaly detection had 92% threat detection rate with 8% false positive rate. Simpler to the zero-day threat study, AI threat detection was superior to human performance and traditional rule-based systems.

The AI threat detection system has flaws as the AI needs to be trained and learn overtime to comprehend potential threats to maintain defence over evolving threats. A substantial flaw is Adversarial Machine Learning (AML) which was researched by Szegedy (Goodfellow, Shlens and Szegedy, 2014). Discovered that some machine learning models are vulnerable to adversarial machine learning, which is an act of manipulating data or the machine's logic resulting in misclassification or failing leading to incorrect predictions. This is challenging due to the significance of adversarial AI exploits as it attacks the learning process and logic that runs the system, losing Integrity and robustness of applications handling sensitive data.

3.3 Findings

Security specialists using machine learning and AI threat detection systems need understanding of AML as AML can cause backdoor attacks, AI models and machine learning should be considered as a method of defence from threats but as a potential threat of exploitation. The complication of using AI and machine learning is identifying how the models learn and what algorithms are used to choose the input and output. If the system has prediction errors or pattern recognition the model can be vulnerable to adversaries exploit by crafting data to the model’s decision tree manipulating it to have incorrect decisions resulting in a security risk.

3.4 Adversarial exploit

An example of an adversarial exploit was in September 2025 when attackers used an AI language model called Claude AI developed by Anthropic. (*Disrupting the first reported AI-orchestrated cyber espionage campaign*, 2025). The threat attacker manipulated Claude AI code tool to execute cyberattacks by itself. The AI targeted tech companies involving financial institutions and government agencies, the AI had success with four confirmed breaches in which the AI had tasks to credential harvest and extract sensitive data such as privilege accounts belonging to the company.

The attacker was able to bypass the AI that was trained to avoid harmful behaviour to engage in the attack by breaking down the attack into small tasks that Claude AI would execute without providing the context of the task being malicious. The reason the AI followed the interactions set by the attacker was due to the attacker stating, “it was an employee of a legitimate cybersecurity firm and was being used in defensive testing”. Claude AI identified and tested vulnerabilities in the targeted organization’s systems allowing for the attacker to use the AI to excuse the attack and provide a document of the attack with the data from the attack. This outlines that AI models can be used for cyberattacks and developers need to be cautious of how their AI

models can be manipulated into doing malicious actions that disregards their guidelines showing that adversarial exploits can happen.

National Cyber Security Centre (NCSC, 2025). stated by 2027 “AI cyber capability is likely to make cyber security at scale increasingly important to 2027 and beyond”. As AI systems are learning constantly to look for vulnerability and exploits it's clear that AI threat detection is the future of cyber security although the risks of AI tools will be enhanced by bad actors using tools for exploits knowing vulnerabilities in AI systems will increase due to AI threat detection systems becoming the standard. NCSC also stated users have already found issues in their systems and are mitigating the vulnerabilities before bad actors can exploit the vulnerabilities. AI will increase the rate of exploitation then prior to before due to advancements in AI which will increase threats due to attackers using AI for attacks.

Research Gaps

1. Transparency and Ethical

From the research I've done from the literature review, I've found that the transparency and ethical use of using artificial intelligence is still developing with no guidance allowing for the use of AI in businesses to use AI unethical and not stating the transparency of the AI model such as 2wai and AI applications making decisions that should be fair to prevent discrimination. There are gaps within the transparency and ethical use of artificial intelligence as businesses and AI systems are being used unethically and not disclosing the transparency of the AI system. Resulting in AI systems not being accountable for their actions and causing unfairness or being biassed to different individuals. Businesses are using AI unethically without considering the impact it could result to the users, disregarding the harm the AI model could make with its decision making.

2. Legal framework guidelines

I found that there are few legal framework guidelines for artificial intelligence and only one dedicated legal framework by the European Union called EU AI act to ensure a safe, transparent and non-discriminatory environment using AI and AI systems should be supervised by humans rather than automation to prevent harmful outcomes. I found that the United States doesn't have a legal framework for AI. Instead, they are either using laws that are already in place or state laws that have AI laws in place. It's clear there are gaps within the legal framework as the guidelines are still developing or don’t have a dedicated framework for artificial intelligence due to complications with the Black Box problem as understanding how AI makes

decisions with the input given to the machine to make the output is unclear due to the lack of algorithmic transparency as the process of making decision cannot be explained. This results in harming the user or being manipulated by the AI due to the lack of clarity in enforcing universal guidelines for artificial intelligence models and systems.

3. AI threat detection

I found that AI threat detection is effective and is becoming stranded in cyber security as AI threat detection are more accurate than traditional threat detection methods and performance of AI and human performance in detecting zero-day threats showed that AI detention accuracy performed 85% compared to human performed 60% although AI systems can be compromised by Adversarial machine learning causing manipulating data or misclassify threats resulting in attackers using that as a vulnerability to use the AI system to their advantage for cyber-attacks.

There are a few gaps with AI threat detection as it's clear that AI threat detection is more accurate than human detection as it operates constantly. There is a threat that machine learning models are vulnerable to adversarial machine learning where attackers can manipulate the machine's logic causing misclassification of incorrect predications. Attackers can use AI for cyber-attacks by manipulating the AI system to cause harm.

4. Methodology

Research design

My research project will follow a structure to ensure the research and predictive analysis has the desired outcome. The data I will be collecting is exclusively secondary data such as reports, published studies in the field and analyses. The research design I will be using is a systematic approach allowing me to collect secondary data to identify gaps and the ongoing advancements in the field of artificial intelligence to make a qualitative synthesis approach. My data sources and selection approach is to gather high standard secondary data to cover my objectives regarding ethical issues, legal guidelines and threat detection. I will be adopting a thematic synthesis approach (Thomas & Harden, 2008) as it will integrate findings

into a report on the potential of AI while addressing concerns of ethical, legal and threat detection.

Using secondary data eliminates ethical issues with primary data such as human participants or privacy concerns regarding surveys related to artificial intelligence. My project will adopt the PRISMA which is the Preferred Reporting Items for systematic Reviews. The reason for this approach is to ensure I follow the guidelines for transparent reporting.

Data sources

The data sources used in this project are summarised in Table 1. Data will be gathered from research papers and using academic databases such as Met search or Google scholar. Gathering legal information and frameworks I will be using government sources such as EU reports for AI policy and ethical transparency.

All sources will be relevant to artificial intelligence and will address AI's future and trends using AI, recency of data that has been published for analysis will be post 2018 with some expectations to clarify AI's developments. Some sources will be grey literature such as papers from tech firms involving artificial intelligence such as OpenAI and media analyses on AI models to provide me with information about the AI model's decision making and transparency. Preference will be given to material published after 2018 (with a small number of earlier exceptions where needed to clarify key developments). Grey literature, including technical reports and whitepapers from AI companies and media analyses of AI models, will also be used to examine decision-making processes and transparency. As shown in Table 1, academic sources will be drawn from MET search, Scopus, and IEEE Xplore, while policy and grey literature sources include EU AI Act reports, UK government reports, US AI papers, Anthropic reports, and OpenAI whitepapers.

Table 1: Data sources

Category	Research Databases
Academic	Met search, Google Scholar, Scopus and IEEE Xplore
Policy and legal information (Gray)	EU AI Act reports, UK government reports and US AI papers
(Grey literature)	Anthropic reports and OpenAI whitepapers. Google advance search to specify searches

Inclusion Criteria and Exclusion Criteria

The relevance of the sources must cover my aim and objectives particularly the ethical use of AI, legal framework and threat detection. Sources that do not directly address these areas will be excluded. All sources will be peer-reviewed or high-quality grey literature with credible citations. As shown in Table 2, inclusion and exclusion criteria will ensure the final dataset consists of recent, relevant, and high-standard material.

Table 2: Inclusion and exclusion criteria

Inclusion	Exclusion
Peer-reviewed papers or high-quality grey literature with credible sources. Whitepapers with citations	Opinions and uncited blogs
Published from 2018 to present to capture the growth of artificial intelligence	papers pre 2018 that are cited to analyses for historical context

Using grey literature sources, will require additional scrutiny to identify if they hold the same standards then research papers while taking accountability that sources may include biassed data or information. Sources that are clearly outdated which are not relevant to artificial intelligence development will not be included. Any sources that are blog posts that don't have citations don't meet the standards of my findings philosophy.

These guidelines will ensure the research collected matches my criteria, while collecting sources I will be aiming for 50 to 70 sources initially to gather as many resources as possible, after peer reviewing the sources I will refine to 20 to 50 sources that fully align with the objectives and meet the required standard. This approach will provide a focused set of high-quality papers while retaining awareness of the relevancies.

5. Artefact Development

Research found regarding the use of AI in today's society and how its development from 1955 to present time is remarkable as it went from a concept of using tasks on computers to understand human level intelligence to where it is now. I will now conduct a theoretical approach with the research collected from my literature review and give theoretical solutions to the gaps within artificial intelligence.

In 2024, Stanford University reported over 78% of organisations have used AI in real-world applications. (Stanford University, 2025). The issue with organizations using artificial intelligence in some capacity whether its generative AI or any AI use is the transparency and ethical considerations while implementing the use of AI into an organization. As stated in my Literature review, cases exist of organisations using AI and not declaring transparency and ethical use resulting in organisations not taking accountability for their actions or the AI they employ. Numerous reasons why organizations struggle addressing the use of AI transparency and ethics such as the Black Box problem and algorithmic bias such as the case in my literature review with a healthcare system, which used an AI algorithm that was favouriting white patients over other races leading to discrimination.

Businesses outside of the EU that are using artificial intelligence need to make a framework to follow since they are only obligated to follow the EU AI Act if they serve EU customers even if they don't have a presence in the EU. The US Federal Guidance doesn't have a comprehensive AI law which is an issue due to the US being the leading production in AI models.

I propose businesses in The United States of America should follow a framework such as this:

5.1 Framework proposal for The United States of America

This framework is designed to establish a national standard for the responsible development, deployment and guidance of Artificial intelligence (AI) systems in the United States of America.

The core aims are:

- Prevent harm to individuals caused by AI systems using driven decisions.
- Mandate transparency in AI systems, including disclose how the decision output was made by including data sources that the AI used and algorithmic methods.
- Mitigate risks of bias discrimination.
- Ensure AI technology aligns with ethical principles, including privacy, accountability, and safety.

- Establish obligations for businesses and developers operating AI systems in the United States of America.

5.2 Transparency and Disclosure

All AI systems must disclose how decision outputs are made, including:

- The data used to train or inform the AI system.
- The algorithms and models employed in generating outputs.
- Whether AI was used in producing content, recommendations, or decisions presented to users.

Businesses must be obliged to inform users and customers when AI systems are being used in processes that affect the customers or users.

- ### Ethical Standards
- AI systems must be accountable for their outputs and operate in compliance with applicable data protection regulations.
- AI systems should be designed to be fair and mitigating bias outputs.
- Human oversight is mandatory to ensure the AI is compiling with the standards.

Documentation and Reporting

- Developers must have detailed records of the AI system design and intended purpose. Records of testing and results of outputs.
- Detailed reports of regulatory disclosures for AI systems that are considered high-risk.

5.3. Risk Classification

Businesses using AI systems must conduct a Risk Assessment to classify each system according to Table 3 below. Classification determines applicable obligations.

Table 3 – Risk Classification with example use cases

Risk Tier	Description	Example Use Cases

Unacceptable Risk	AI that directs harm, exploits vulnerabilities, or violates fundamental rights to the user.	Manipulative behaviour targeting unauthorized biometric surveillance targeting minors under 13.
High Risk	AI operating in sensitive domains with significant impact on individuals' rights or welfare.	Healthcare diagnostics; hiring and credit scoring; public safety systems.
General Risk	AI using personal data or sensitive outputs that has potential for misuse.	Collecting user data without permission

Unacceptable Risk

- Unacceptable Risk: AI systems that can cause harm to users including manipulative techniques and exploitative to user's vulnerability.
- AI systems using real-time biometric identification without consent from users and customers with exceptions to authorities for crimes.
- Market research users without consent and on children under the age of 13 restricts on using services.

High Risk

- AI systems that are being used in sensitive departments within an organization require audits for bias and safeguarding against discrimination.

General Risk:

- Unauthorized use of customers' sensitive data and personal data in training AI and dissemination of harmful outputs.

5.4 Obligations for Transparency and Ethics:

All businesses deploying AI systems must:

- Offer privacy control for minor protection, clear reporting systems in place.
- Clear transparency disclosures on artificial generated content.
- Consent must be permitted for sensitive data being used in training.

Ethical Implementation

- Businesses must integrate safeguarding policy to mitigate physical, financial harm and psychological harm.
- Mandate design features that can cause exploitation.

High-Risk

- Mandatory risk assessments and impact evaluations to prevent user discrimination.
- Developers must ensure ethics training for users and provide reporting systems for safety and concerns.

5.5. Governance and Regulations

Institutions

- National Institute of Standards and Technology (NIST) for evaluating risks and safety testing.

- Follow ethical governance standards with the national Artificial Intelligence Research Resource (NAIRR) for guidance.
- The Department of Homeland Security should supervise risks of artificial intelligence that can cause damage to the country.

Stakeholder Involvement

- Businesses should allow input from stakeholders for standards when establishing internal AI standards.

- Stakeholders must be consulted in agreements and policy changes in the development of AI and changes.

State Laws Pre-emption

- AI laws permitted from the U.S. Congress should be followed by all States in America.

- AI State laws can enforce any changes made after discussing with U.S. Congress.

Monitoring

- Annual assessments and audits required by Office of Science and Technology (OSTP).

- Platforms must apply to maintain data protections and audit requirements.

Key Definitions

Table 4 – Key Definitions

Term	Definition
AI System	Any technological machine system that generates output such as decisions, recommendations, or content based on the input given from the user.

High-Risk AI	An AI system that handles sensitive information or a sensitive domain where the output is at risk of affecting individuals' safety or rights.
Biometric Data	Personal data from technical processing of physical recognitions or behavioural characteristics that can define a person.
Minor	Person that is under the legal age of 18 in the state.
Sensitive Data	Sensitive information such as financial information, ethnic origin, health status or religious beliefs.
Bias Audit	Evaluation of an AI system’s data and outputs to document and identify patterns of discriminatory or unacceptable outputs.

Rationale of the Proposed Framework

This framework was inspired by the EU AI protection Act while taking into consideration regulations already in place by the US. By applying this framework in the USA, it sets a standard and regulations addressing issues and concerns with ethics and transparency in AI systems across the US by leveraging existing agencies such as the NIST for technical standards and NAIRR for ethical guidance for distributing regulatory authority oversight that businesses can follow and compliance with the framework. This will give businesses in the US guidance to be accountable for their actions using AI systems to ensure transparency, ethics, bias, discrimination and declaring the use of AI meet the requirements of the framework.

This framework isn't intended to restrict artificial intelligence and innovation for new AI applications or reduce funding for AI research in the US; rather establish potential risks that AI systems and models can pose against customers, users and the public.

The reasoning of bias audits in high-risk AI systems (Section 5.3) is demonstrated by the 2019 Optum healthcare algorithm case where an AI system exhibited racial bias favouring white patients over Black patients with identical health conditions (Hickok et al.2022). That case illustrates that AI systems can have bias outputs through natural variables which requires bias audits not including standard risk assessments, if Optum followed the proposed framework and demonstrated bias audits in bias detection and mitigation it could have mandated AI systems to be unbiased to patients of all backgrounds where algorithmic decisions directly impacts patient care and resources given to them.

The EI AI Act's risk-based classification system (unacceptable, high, general risk) provides a proven model for AI regulation that is essential for risk management that the proposed US framework adapts to American regulation structures. The EU AI Act prohibition of AI applications that can manipulate users such as vulnerable groups which directly addresses cases like 2wai, which monetizes grief by creating AI avatars of deceased individuals in which an application of that nature would be banned under EU law but currently operates legally in the United States. The proposed US framework classifies manipulative AI as Unacceptable Risk and market research on children under the age of 13.

6. Delay in producing Legal and Regulatory Frameworks for Artificial Intelligence

Artificial intelligence has been around since the 1950s with scientists and inventors developing first generation AI applications and machines with no limitations of ethical and legal issues. Scientists and inventors just saw issues for example Richard E. Susskind and Philip Capper developed a fully operational AI systems lawyers to use called the "Latent Damage Law – The Expert System", which they created a decision tree with over two million paths to predict the likely outcome including permutations of fact patterns and legal problems.(Susskind, 1989) The purpose of this application was to build a system that reduces lawyers research time. Susskind did come across legal issues as who would be liable for using proposed system legal advice that made errors in law. Even in 1989 inventors were considering the issues that could occur in the foreseeable future.

I believe the reason scientists and inventors didn't consider legal implications with artificial intelligence at the time was due to AI applications and machines being used in a controlled environment such as research labs or areas where the impact was limited rather than societal integration. Scientists and inventors were more focused on the theoretical and challenges with achieving machine learning and identifying patterns with AI rather than considering real world anticipations such as bias or discriminatory outcomes. The rate of technology advanced significantly to a rate humans didn't consider the thought of AI being an imminent threat to humanity. Even after the second generation of AI systems got created it didn't seem like a concern as the less technical AI systems and models did not have the capability to be threatening in the foreseeable future, developers and researchers in the field of

artificial intelligence failed to see that AI systems would outperform humans in a variety of tasks even when there was proof of AI systems learning patterns to eventually outperform humans such as Deep Blue computer in 1997 which wasn't classed as second generation of AI. The main takeaway to be learned from the past is to not underestimate the pace of how technology can develop rapidly and not to dismiss the threats that can occur, be prepared for the unthinkable.

Evaluation of the Artefact

1. Proposed Framework Development

The structure compounds cover Risk-based classification with three tiers to determine what AI systems fall under, with institutional oversight mechanisms in place to guide AI systems and clear definitional framework.

1.1. Evidence-based design

Bias Audit requirements (Section 5.2): Directly correlates by Optum case demonstrating the need for specialized detection in place.

Transparency Mandates (Section 5.2): Directly justified by the Black Box problem and accountability gaps.

Unacceptable Risk Category (Section 5.3): Directly correlates by 2wai and Character.AI case showing exploitation of vulnerable groups.

Institutional Architecture: Capitalizes on existing US agencies instead of creating new bureaucracy.

1.2. Innovation preserving approach

The framework is explicitly designed not to restrict AI innovation and maintains research funding support, focusing on risk mitigation rather than prohibition approach.

The framework adapts the proven EU framework to American regulatory culture such as distributed authority and centralized oversight.

1.3 Framework validation through cases

The framework adopts a structured ethical and regulatory framework for artificial intelligence. As shown in Table 5 each requirement in the proposed framework

addresses a specific real-world case of AI not being regulated, furthermore if the proposed framework was adopted, the documented harm would be mitigated.

Table 5: AI framework requirements, evidence sources and documented harms

Framework Requirement	Evidence Source	Documented Harm
Bias audits for high-risk AI	Optum healthcare case	Racial discrimination affecting 200 million patients.
Prohibition of manipulative AI	2wai, Character.AI cases	Grief monetization, child grooming and suicide.
Human oversight mandate	Claude AI exploitation	AI manipulation for cyberattacks.
Transparency disclosures	Passport recognition case	Unexplainable automated decisions.
Testing requirements	Threat detection studies	22% potential breach reduction.

2. Contributions to the field

2.1 Integrated Regulatory framework with Evidence Mapping

Unlike prescriptive frameworks, the proposed framework justifies each requirement through documented cases creating traceable links of why the requirements are needed between regulatory provisions and necessity.

Concluding Assessment

This research represents substantive contribution to AI governance discourse through integration of technical performance data, regulatory comparison, ethical case analysis and framework development in an evidence-based argument for comprehensive AI regulation. The Artefact demonstrates how the United States of America can adopt a framework that has rational logic of each requirement in the framework by linking it to a documented case to ensure artificial intelligence is being used ethically.

7. Discussion

While I have my theoretical views about artificial intelligence, researchers and experts in the field have their own views.

A: Regulatory perspective on AI ethics and governance.

B: Data biases, transparency and accountability.

C: Applied perspective with a concrete AI framework for threat detection.

A. International Business Machines Corporation (IBM) stated what AI may look like in 2034 regarding regulations and AI ethics noting AI regulations and ethical standards will have advanced significantly with frameworks such as the EU AI Act being a key development in addressing risks in systems and classifying AI based on risks and strict requirements to follow transparency guidance and standards furthermore mentioning ethical considerations will be included on bans on systems that pose a threat in public spaces and human oversight will be required to address bias and fairness (Mucci, 2024).

B. Another expert in the field stated that data used in training AI systems are not representative of the population or can reflect on historical events as the AI system will likely perpetuate biases. Also, stating the importances of accountability and transparency in AI systems to foster trust and ethical use of the AI systems but due to the complexity of AI systems such as the Black box problem, simpler to the IBM view, this report states the importance of human oversight in AI decision making for enhancing accountability to review AI systems to give a level of judgment and ethical considerations that the AI system can't comprehend. The challenges with many AI systems are based on deep learning through complex algorithms that are hard to interoperable resulting in a lack of transparency making it difficult to regulate on how decisions are made, assess identify potential biases and ethical volitions within AI systems (Kashefi, Kashefi and Ghafouri Mirsaraei, 2024).

C. A theoretical study I found on an approach to threat detection and mitigation is a proposed AI driven framework for predictive threats. By integrating AI in the threat detection process to proactive leverages AI to predict potential threats before they materialize. The effectives of the AI framework depend on the quality of data the AI analyses such as external threats working with government agencies and cyber experts to provide contextual information on attacks, vulnerabilities and adversarial exploits. Also stated that there will be

two different machine learning models, Supervised learning models would use attack types and their behaviour by learning from historical data to accurately predict future attacks. The second model would be an unsupervised learning model for cases where the AI encounters an unknown threat, the model won't rely on labelled data such as historical data instead it would analyse system behaviour to detect anomalies. The study includes a hypothetical study of a financial institution that had traditional methods of defence but wanted to enhance their defence by using AI driven detection for predictive threats including two case scenarios using the AI driven threat detection compared to traditional methods. Traditional methods rely on signature-based detection which is effective if the threat is known which is limited but the proposed AI framework can detect unseen threats based on behavioural anomalies making it more adaptable to new and evolving threats (Hasan et al., 2025).

Comparison between viewpoints

1.Primary Focus

View A focuses on high-level regulatory progress and suggests the EU AI Act as a central framework for development in the future for managing risks and establishing ethical standards including bans on high-risk systems. View B has a simpler approach but states issues on the practical repercussions of non-representative training data and issues with ethical and trust in AI systems due to the black-box nature of AI.

View C primely focuses on technical threat detection with a proposed AI driven framework for predictive threats, leveraging both supervised and unsupervised learning models to adapt to predict and mitigate cyber threats tested using a hypothetical case.

2. Data quality

View A does not clarify on data issues directly while view B explicitly advises that training data can be unrepresentative of historical events which can lead to biases. View C states that the effectiveness and accuracy of the AI threat detection framework depends on the quality of data the AI analyses requiring integration with cyber experts and agencies to give the threat detection system quality data.

3. Human oversight

There is a firm belief on the necessity of human involvement as view A requires for human oversight in public spaces and for AI systems that can pose a threat. View B highlights human oversight is essential for accountability and reviewing outputs of AI decisions, highlighting the complexity of black box systems. View C requires human oversight to supply contextual information related to attacks, vulnerabilities and potential adversarial exploits that the AI can't interpret.

Synthesis Review

While Views A and B cover elements on the need for robust regulatory guidelines, transparency and human oversight to mitigate risks such as ethical and bias related issues in general AI systems, View C demonstrates how principles can be operational in a high-stake domain regarding cyber security. The three perspectives are complementary as governed focused guidelines (A and B) paired with adaptive frameworks such as the proposed AI driven framework (C). There is a clear gap in ethical considerations into AI frameworks as well as embedding transparency and accountability requirements which was highlighted in A and B into systems such as threat detection systems like C.

In comparison with the theoretical analysis set out in the following section for future research to (A) and (B) there is a common ground conception as (A) and (B) views the necessity of human oversight and the integration of fairness and accountability metrics to mitigate bias and ensure ethical measures of deployment. My analysis and both views recognize that transparency and accountability are foundational to ethical safeguarding to regulatory compliance and public trust. My analysis and view (A) acknowledge the need to classify and address risks proactively while my views

and (B) have concerns about real world consequences of underestimating bias and the long-term impacts.

My analysis is prescriptive and advocates for a mandatory pre- release evaluation period on AI systems with a strict separation of impact versus ethical assessments to enforce accountability. View (A) relying on evolving industry wide regulations such as the EU AI Act and advancements by 2034. View (B) focuses on the technical restrictions such as non-representative data and black box opacity that complicates regulation without a specific structural solution like the evaluation period stated in my analysis. My view expresses the uncertainty towards industry behaviour developers and independent developers viewing a competitive pressure which can lead to reckless decisions in developing AI systems. Views (A) and (B) assume that regulatory evolution and transparency standards will improve with human oversight and can address issues with existing development obstacles.

My analysis on threat detection and view C strongly affirm that AI threat detection is the future of security as both views emphasise its ability to address limitations with using traditional methods of handling complex and evolving threats that exceed signature-based approaches. My view and view (C) both highlight the importance of robust data to ensure effective threat identification as well as the value of anomaly detection while my view goes through business monitoring and training and view (C) is a concept of unsupervised models to identify attacks beyond historical patterns.

8. Conclusion

8.1 Key Research Findings

This research identified and documented multiple indications of algorithmic bias and transparency failures such as racial bias towards patients and demonstrated how to reduce 80% of bias reduction is achievable through appropriate auditing. Analysis showed bias can emerge from neutral variables and requires detection further than standard risk assessments. Documented how businesses are using AI to monetize grief through AI avatars and recorded grooming behaviour patterns in AI chatbots leading to documented harm leading to 2/3 of UK children aged 9-17 use chatbots. Analysis showed that AI systems can systematically exploit vulnerable emotional states in applications.

Research analysed existing regulatory frameworks and identified jurisdictional gaps such as the US, despite leading in AI model production lacking comprehensive regulatory standards while EU AI Act establishes human oversight and prohibited applications like manipulative AI. Demonstrated how 2wai application would be prohibited under EU AI Act or proposed US framework, establishing examples where

regulatory gaps permit potentially harmful applications, and regulatory fragmentation creates practical consequences.

Research compiled and analysed comparative performance data showing measurable AI advantages such as Zero-Day Threats, AI 85% vs Human 60% accuracy, efficiency of AI detection speed at 12 minutes vs Human takes 4 hours which speed is critical when attackers exploit vulnerabilities within hours of discovery.

Documented that 43% of UK businesses experience cyber breaches and calculated the theoretical impact if AI detection could reduce breach rate from 43% to 33% which would protect UK businesses annually. Also examined how Adversarial Machine Learning can impact cyber-attacks and stated a real-world exploitation such as the Claude AI manipulation, confirmed four organizational breaches through AI-assisted attacks.

8.2 Future research

1. Ethical AI systems in the future

Developers need to consider the long term impact and purpose of AI capable of the ability to outperform humans doing advance tasks instead of integrating AI into already existing systems and using that as a starting point as they are creating risks of integrating AI intelligence in systems that it wasn’t designed for creating unethical systems, deploying AI systems to the public without considering if the system is ethical or can cause harm. Without regulatory enforcement, companies and independent developers will continue deploying AI systems with minimal safety oversight, creating systemic risk.

2. Is AI Threat Detection the Future?

To answer whether AI Threat Detection is the future, from the research discussed in the literature review and the results from existing data, the evidence strongly supports that AI threat detection is essential for modern cybersecurity infrastructure.

Research findings demonstrate clear performance advantages over human analysts and traditional rule-based systems stated by the research of detection accuracy by Sisodiya (2025) comparing AI systems to human analysts in detecting zero-day threats in which AI achieved 85% detection accuracy compared to 60% for human analysts. Sukara's 2021 study compared AI-based anomaly detection to traditional

rule-based systems and found that AI achieved 92% threat detection rates versus 70% for approaches.

AI threat detection introduces new vulnerabilities such as Adversarial Machine Learning (AML) attacks which exploit AI decision making processes causing misclassification (Szegedy et ai., 2014). The September 2025 Claude exploitation showcased attackers who manipulated Anthropic's Claude AI to aid a malicious task successfully breaching four organisations. This case illustrates that AI improves detection of traditional threats but creates attack scenarios that didn't exist with rule-based systems.

3. Future of Regulatory of AI

When addressing the legal and regulatory framework for artificial intelligence the clear answer may seem to regulate and control artificial intelligence, ensuring only trustworthy AI, safe AI or responsible AI is used. This is not the answer as with most technological inventions, the objective is to enhance and test the capability of the tech such as the advancements, including Smartphones, Computers and the Internet of Things (IoT). These technologies do not stop production or have a definitive goal they simply carry-on improving overtime and Artificial intelligence and Machine Learning is more transformative than the tech listed.

Recommendations

1. A beneficial proposal for regulating AI would be to have a mandatory evaluation period till the AI model is pushed in public domains or professional environments. During the timescale developers are under strict authority to underline the goal of the AI and conduct a comprehensive assessment, if the impact of the risk is significant then it shouldn't be distributed at all. The reason for this is we don't want history to repeat itself by underestimating the impact of the risk and causing harm to the user or disrupting the work environment.

Risk assessment concerning the impact should be structurally separated from evaluations of ethical and bias of AI systems and should have a specialized assessment to gain a deep analysis and accountability without technical feasibility considerations. While conducting the assessment the metric of explicitly identifying bias and ethical issues should include fairness and accountability are essential it meets regulatory compliance and transparency. Separating impact assessment from ethical-bias evaluation produces a more precise and actionable understanding.

This proposal faces challenges as developers may be too reckless to understand the impact of the risks, it doesn't help that major corporations (Microsoft, Google, Nvidia) prioritize speed over safety, racing to dominate AI markets. This competitive pressure causes developers to dismiss long-term risks, integrating AI into systems it wasn't designed for without proper safety consideration.

2. Implementing AI threat detection is crucial in the climate of Cyber Security given that in 2025 alone in the UK 43% of businesses reported they experienced a cyber security breach or attack on the business. (Government of the UK, 2025) If AI threat detection can identify 92% of threats versus 70% for traditional methods, implementing AI could theoretically prevent 22% if successful attacks potentially reducing the UK breach rate from 43% to approximately 33% which would protect thousands of businesses in the UK annually.

Constant testing must be in place to ensure AI can detect patterns of anomalies in the business to stop false flagging and advantage penetration testing against the AI threat detection system to give the AI data that it can use against potential security attacks. If the AI threat detection system doesn't have data on every possible attack vector that could occur on the business, the threat detection systems will be inaccurate or not detect the threat due to not having data on the attack vector and not having a solution to mitigate the attack. It is critical if businesses are relying on AI threat detection systems then they should take time developing the detection system to a set standards by penetration testers as it can be time consuming training the AI threat detection systems on possible threats and what are the solutions to mitigate the attacks and also letting the AI detect anomalies in the business.

AI threat detection is superior for known and emerging threat patterns but not as a standalone solution to threat detection as creating an optimal architecture combining AI detection systems with human oversight to address adversarial attacks and threat vectors in the future that threat detection systems don't have training data on.

3. All AI systems will need to be regulated in the current climate and future AI systems to ensure the safety and compliance with regulations. Major evaluations regarding legal and regulatory frameworks need to be in place every two to three

years compared to three to five years which is usually the standard for reviewing frameworks. As previously stated, artificial intelligence systems can evolve rapidly often in a couple of months. The existing regulatory framework fails to keep pace with technological advancements, posing significant risks to the safety and responsible deployment of AI systems. Legal and regulatory frameworks governing AI must remain continuously updated. The inherent uncertainty surrounding the future capabilities of AI, ongoing adaptation of these frameworks is essential to mitigate potential risks, including algorithmic bias and harmful outputs.

Research Contribution

1. These cases demonstrate the Black Box problem creates accountability gaps with measurable harm. The research establishes that in the current market, driven approaches to AI ethics are insufficient as evidenced by the Optum algorithm used on patients before bias detection through academic research rather than internal testing before being deployed.
2. The analysis demonstrates that the EU's risk-based approach provides a regulatory model to follow while US fragmentation creates gaps that can be exploited, therefore it's critical that the US follows a framework such as the proposed framework to determine whether harmful applications can operate.
3. The research answered the question "Is AI Threat Detection the future?" with evidence that demonstrates that AI threat detection shows superiority for pattern detection and known threats while identifying that architecture requires AI and human collaboration to address threats that AI can overlook.

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